

**HARAMAYA UNIVERSITY**  
**COLLEGE OF BUSINESS AND ECONOMICS**  
**DEPARTMENT OF ECONOMIC**  
**MOCK EXAMINATION**

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**MACROECONOMICS REVIEW QUESTIONS**

**Multiple-Choice Questions: Choose the best answer and round it (1 pt. each)**

1. (Advanced analysis) The equation  $C = 35 + 0.75Y$ , where C is consumption and Y is disposable income, shows that:
  - A. Households will consume three-fourths of whatever level of disposable income they receive.
  - B. Households will consume ETB 35 if their disposable income is zero and will consume three-fourths of any increase in disposable income they receive.
  - C. There is an inverse relationship between disposable income and consumption.
  - D. Households will save ETB 35 if their disposable income is zero and will consume three-fourths of any increase in disposable income they receive.
2. (Advanced analysis) If the equation for the consumption schedule is  $C = 20 + 0.8Y$ , where C is consumption and Y is disposable income, then the average propensity to consume is 1 when disposable income is:
  - A. ETB 80.
  - B. ETB 100.
  - C. ETB 120.
  - D. ETB 160.
3. Which consumption theories explain the fact that a person will consume more if he is living in a community of high income than if he lives in a community of low income
  - A. Demonstration effect theory
  - B. Permanent income hypothesis
  - C. Life cycle hypothesis
  - D. Intertemporal theory of consumption
4. According to Modigliani Life Cycle Hypothesis, which of the following pairs of income pattern and income related and consumption activities are paired incorrectly
  - A. Young age low income – borrowing
  - B. Middle age high income – dis-saving

- C. Old age low income – dis-saving
  - D. Old age low income – borrowing
5. (Advanced analysis) Answer the next question(s) on the basis of the following consumption schedule:  $C = 20 + .9Y$ , where C is consumption and Y is disposable income. At an ETB 800 level of disposable income, the level of saving is:
    - A. ETB 180.
    - B. ETB 740.
    - C. ETB 60.
    - D. ET B18.
  6. Holly's break-even level of income is ETB 10,000 and her MPC is 0.75. If her actual disposable income is ETB 16,000, her level of:
    - A. Consumption spending will be ETB 14,500.
    - B. Consumption spending will be ETB 13,000.
    - C. Consumption spending will be ETB 15,500.
    - D. Saving will be ETB 2,500.
  7. If government increases lump-sum taxes by ETB 20 billion and the economy's MPS is .4, then the:
    - A. Consumption schedule will shift upward by ETB 12 billion.
    - B. Consumption schedule will shift downward by ETB 12 billion.
    - C. Equilibrium GDP will increase by ETB 8 billion.
    - D. Equilibrium GDP will decrease by ETB 8 billion.
  8. Which of the following money demand is **NOT** related to interest rate?
    - A. Speculative money demand
    - B. Transaction money demand
    - C. Portfolio money demand
    - D. All of the above
  9. Which of the following would reduce the money supply
    - A. Commercial banks use excess reserves to buy government bonds from the public.
    - B. Commercial banks loan out excess reserves.
    - C. Commercial banks sell government bonds to the public.
    - D. A check clears from Bank A to Bank B.
  10. Business cycle theory which states changes in aggregate spending is the cause of variations in real GDP of an economy is known as:
    - A. The monetarist theory

- B. Rational expectation theory
  - C. The Keynesian theory
  - D. Theory Political Business Cycle
11. High government investment may result ..... on the private sector investment,
- A. A crowding out effect
  - B. Paradox of thrift
  - C. Crowding in effect
  - D. Crowding between effect
12. High savings may result in a malfunctioning effect on the economy, it's called
- A. A crowding out effect
  - B. Paradox of thrift
  - C. Discretion effect
  - D. Pulling in effect

**Use the following to answer questions 1-3:**

Answer the next question(s) on the basis of the following information about the hypothetical economy of ETHIOPIA.

**All figures are in millions.**

|                         |            |
|-------------------------|------------|
| Not in the labor force: | <b>45</b>  |
| Unemployed:             | <b>7</b>   |
| Total population:       | <b>145</b> |
| Employed:               | <b>95</b>  |
| Discouraged workers:    | <b>3</b>   |

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13. Refer to the above information. The labor force in ETHIOPIA is:
- A. 95 million.
  - B. 102 million.
  - C. 105 million.
  - D. 145 million.
14. Refer to the above information. The unemployment rate in ETHIOPIA is:
- A. 2.5 percent.
  - B. 3.2 percent.
  - C. 5.0 percent.
  - D. 6.9 percent.

15. Refer to the above information. If the natural rate of unemployment in ETHIOPIA is 5 percent, then:
- A. Structural unemployment is 3 percent.
  - B. Cyclical unemployment is 2 percent.
  - C. Frictional unemployment is 2 percent.
  - D. Hidden unemployment is 5 percent.
16. Wrong statement about Gross domestic product is \_\_\_\_\_
- A. Total income of factors in an economy in a year
  - B. Total expenditure on the economy's output of final goods and services
  - C. Value added production in an economy in a year
  - D. Total income plus total expenditure in an economy
  - E. None
17. The function of money to transfer the purchasing power from present to future is \_\_\_\_
- A. Money as unit of account
  - B. Money as store of value
  - C. Money as medium of exchange
  - D. All
18. Suppose that you saved some amount of money in the bank at 7% annual interest for one year. If inflation in that year was 10%, which of the following statements is correct?
- A. You are richer 7% than you were
  - B. Nominal interest rate is 3%
  - C. Real interest rate is 17%
  - D. Purchasing power of your money declined by 3%
19. Which is right for an open economy model?
- A. Policies that tend to increase saving cause trade deficit
  - B. Policies that tend to decrease investment cause trade surplus
  - C. Policies that tend to increase investment and decrease saving cause trade deficit
  - D. None
20. Suppose that 100 tons of wheat sold in a year at 5\$ per ton of wheat. If total money supply is 20\$, what is velocity of money?
- A. 60
  - B. 40
  - C. 25
  - D. 20

E. None

#### MICROECONOMICS REVIEW QUESTIONS

#### Multiple-Choice Questions: Choose the best answer and round it (1 pt each)

1. Let good X and good Y is Substitutes. When the price of good Y decreases,
  - A. The demand good Y will decrease
  - B. The demand of good X will increase
  - C. The demand of good X will decrease
  - D. The demand of good X will not change
2. Which of the following determinants can increase the demand for *Teff*?
  - A. Decrease in the price of wheat
  - B. Increase in the population
  - C. Increase in the price of *Teff*
  - D. Decrease in the income of population
  - E. All
3. An increasing in demand and a decrease in supply will \_\_\_\_\_price and \_\_\_\_\_ Quantity.
  - A. Decrease/ Increase
  - B. Indeterminate effect on /increase
  - C. Increase / indeterminate effect on
  - D. Increase/ decrease
  - E. Increase/ increase
4. Perfectively competition is characterized by:
  - A. Large numbers of firms, heterogeneous product, and easy entry into and exit.
  - B. Large numbers of firms, homogenous product and incomplete information
  - C. Large numbers of firms, homogenous product, and easy entry and exit.
  - D. Few firms, differentiated product, easy entry and exit from the market.
5. The relevant stage of production of a rational firm is between
  - A. Maximum MP & Maximum AP
  - B. Maximum AP & Maximum TP
  - C. Maximum MP & Maximum TP
  - D. Maximum TP & zero MP
6. Which of the following is/ are fixed cost

- A. Cost of labour
  - B. Cost of fuel
  - C. C. Interest on borrowed funds
  - D. None
7. The quantity demand of good C has increased from its initial amount to 75 units, as good D's price changed from 5 to 10. Find the initial amount of quantity if the cross elasticity is 2.
- A. 40
  - B. 35
  - C. 25
  - D. 38
8. All are true about consumer's equilibrium except;
- A. At equilibrium MRS equals the slope of budget line
  - B. At equilibrium the slope of the indifference curve is equal to the ratio of the price
  - C. Budget line is equal to indifference curve
  - D. A&C
9. Suppose the wheat farmer producing in perfectly competitive market sells one KG by 8 birr. If the total cost of production is given by
- $$TC = 10Q^3 - 250Q^2 - 1992Q + 60,000$$
- What is the profit maximizing level of output for the farmer?
- A. 14
  - B. 25
  - C. 31.25
  - D. 5
  - E. None
10. Depending up on question "9", what will be the level of profit at equilibrium
- A. 180
  - B. 1492.8
  - C. 0
  - D. -323
  - E. None
11. Which of the following is true?
- A. Nash equilibrium is the special case of dominant strategy.
  - B. A game must have a single Nash equilibrium.
  - C. In some game the dominant strategy may be absent.

- D. Under Nash equilibrium individual will maximize own benefit irrespective of other.
12. Which of the following is the property of public goods?
- A. Transferability
  - B. Inclusive
  - C. Non-rival
  - D. B and C
13. Suppose **Muger Cement Factory** is producing cement in perfectly competitive factor and product market. The factory uses labour as a variable factor of production and  $VMP_L$  and its  $MC_L$  is given by:-
- $VMP_L = -14L + 960$                        $VMP_L$ , value of marginal product of labour  
 $MC_L = 120$                                        $MC_L$ , marginal cost of labour
- What is the profit maximizing level of labour employment based on the above information?
- A. 56
  - B. 31
  - C. 60
  - D. 97
14. Which of the following variable is negatively related with demand for labour?
- A. Price of output
  - B. Marginal productivity of labour
  - C. Wage of workers
  - D. B and C
15. Market failure is exist when;
- A. The individual interest is equitable with social interests
  - B. The market is perfectly competitive
  - C. The price fails to show exact benefit and cost of production.
  - D. A and C
16. The supply of labor by an individual worker will be backward bending when
- A. The Substitution effect is dominating the Income effect
  - B. The Income effect is dominating the Substitution effect
  - C. The Income effect is equivalent to the Substitution effect
  - D. None
17. Why  $VMP_L$  is down ward sloping in perfectly competitive market structure?
- A. Because the price of the commodity can be decrease.
  - B. Because the additional unit of labor employed will produce more of output

- C. Because when Wage is greater than VMPL the individual should hire more of labor.
- D. Because of law of diminishing marginal productivity.
18. Which of the following is the property of the oligopoly market?
- A. There are large number of producers and few buyers
- B. Products can be differentiated
- C. Producers are independent
- D. There will be free entry and exit
19. How much output each firm will produce if there are four oligopoly firms in the industry?  
In case of Cournot having the same cost and demand information.
- A. 0.25
- B. 0.5
- C. 0.2
- D. None
20. Assume firm B is a sophisticated firm and A is a follower firm, which of the following is true about Stackelberg's duopoly model?
- A. Firm A produces on B's reaction curve
- B. Both will produce on their own reaction curve
- C. Firm A will get the maximum possible profit given the output of B
- D. Firm B will get the maximum possible profit given the output of A
- E. A & B
- F. All except B

## ECONOMETRICS

### Multiple-Choice Questions: Choose the Correct Answer

1. Suppose that an undergraduate student has used the following multiple linear regression model to investigate the determinants of students' performance in Haramaya University by employing cumulative GPA ( $CGPA_i$ ) as the dependent variable as follows.

$$CGPA_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + U_i$$

Where  $CGPA$  is the cumulative GPA,  $X_1$  is the income of the parent of the student and  $X_2$  is the time spent on studying. If  $X_2 = 0.5X_1$ , what assumption of the linear regression model would be violated?

- A. The assumption of no autocorrelation
- B. The assumption of no perfect multicollinearity

- C. The assumption of homoscedasticity
  - D. The assumption of constant variance of  $U_i$
1. In the regression model  $Y_i = \beta_0 + \beta_1 DC_i + \beta_2 DF_i + \beta_3 (DC_i \times DF_i) + u_i$ , where  $Y$  denotes earnings,  $DC$  is a dummy variable for having a college degree, and  $DF$  is a gender dummy variable,  $\beta_2$ 
    - A. is the gender difference in earnings for someone with a college degree.
    - B. is the gender difference in earnings for someone without a college degree.
    - C. is the differential intercept of gender for those with and without a college degree
    - D. cannot be estimated since  $F_i$  and  $(C_i \times F_i)$  are perfectly collinear when  $F_i = 0$ .
  2. The proportion of variability in the dependent variable that is explained by the independent variable in the regression model is now known as
    - A. Coefficient of correlation
    - B. Coefficient of regression
    - C. Coefficient of determination
    - D. Coefficient of interaction
  3. Why is the number of dummy variables to be entered into the regression model always equal to the number of groups ( $k$ ) minus 1 ( $k-1$ )?
    - A. To avoid the problem of serial correlation
    - B. To increase the R-squared value
    - C. To avoid the situation of perfect multicollinearity
    - D. To control for other variables in the model
  4. Given the coefficient of determination or  $R^2 = 0.24$ , sample size or  $n = 25$ , and the number of parameters or  $k = 2$ , the adjusted- coefficient of determination will be
    - A. 0.034
    - B. 0.030
    - C. 0.178
    - D. 0.781
  5. The group, category, or classification that is assigned with the value of 0 is referred to as \_\_\_\_\_.
    - A. the base category
    - B. the benchmark category
    - C. the control category
    - D. the omitted category

- E. All
6. Time series analysis helps to:
- A. Understand the behaviors of a variable in the past
  - B. Predict the future behaviors of a variable
  - C. Plan future operations
  - D. All of the above
7. How do we interpret a dummy variable coefficient?
- A. The difference between the two means
  - B. The difference between the two coefficients
  - C. The difference between two R-square values
  - D. None of the above
8. Heteroscedasticity is more likely in a problem of
- A. Time series data
  - B. Pooled data
  - C. Cross-sectional data
  - D. A & C
9. Panel data have several advantages over purely cross-sectional or purely time series data. These include:
- A. Increase in the sample size
  - B. Study of dynamic changes in cross-sectional units over time
  - C. Study of more complicated behavioral models
  - D. All
  - E. None
10. Which one of the following is not among the consequences of Heteroscedasticity
- A. The OLS estimators will have a bias
  - B. The variance of OLS coefficients will be incorrect
  - C. OLS estimators will be inefficient
  - D. None

Suppose that there is common sense that money demand depends on the interest rate and real income. Based on this idea you have collected sample data presented in the following table. Then answer the questions below the table (questions 12 - 15).

|                    |    |    |    |    |    |    |    |    |    |    |
|--------------------|----|----|----|----|----|----|----|----|----|----|
| Money Demand       | 21 | 16 | 18 | 25 | 18 | 43 | 44 | 42 | 32 | 36 |
| Real interest rate | 3  | 5  | 3  | 4  | 3  | 6  | 12 | 8  | 7  | 7  |
| Real income        | 25 | 26 | 21 | 23 | 24 | 36 | 45 | 36 | 35 | 37 |

Let  $Y$  = Money Demand;  $X_1$  = Real interest rate and  $X_2$  = Real income

$$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + e_i$$

11. Given the above data the estimated value of  $\beta_1$  = \_\_\_\_\_
  - A. 22.5
  - B. 1.699
  - C. -1.236
  - D. -15.67
12. Given the above data the estimated value of  $\beta_2$  = \_\_\_\_\_
  - A. 22.5
  - B. 1.699
  - C. -1.236
  - D. -15.67
13. Given the above data the estimated value of  $\alpha$  = \_\_\_\_\_
  - A. 22.5
  - B. 1.699
  - C. -1.236
  - D. -15.67
14. Which one of the following is not among the reasons for autocorrelation?
  - A. Cyclical fluctuations
  - B. Exclusion of variables from the regression model
  - C. Incorrect functional form of the model
  - D. None of the above
15. \_\_\_\_\_ is the most celebrated test for detecting serial correlation
  - A. Run test
  - B. Goldfeld-Quandt test

- C. Durbin-Watson d test
- D. Park test
- E. None

16. A recognizable set of symptoms for the existence of multicollinearity on which one can rely are

- A. High coefficient of determination
- B. High correlation coefficients
- C. Large standard errors
- D. All of the above

Given the following data on the dependent variable (Y) and independent variable (X), such that Y is modelled as function of X;  $Y = \alpha + \beta_1 X_1 + e_i$ . Then answer the questions below the table (questions 18 - 19).

| No. | Y   | X   |
|-----|-----|-----|
| 1   | 200 | 45  |
| 2   | 300 | 60  |
| 3   | 450 | 75  |
| 4   | 600 | 90  |
| 5   | 800 | 100 |

17. Estimate (compute) the value of the intercept parameter  $\hat{\alpha} =$ \_\_\_\_\_

- A. -313.198
- B. 198.313
- C. -313.313
- D. 198.313

18. Estimate (compute) the value of slope parameter ( $\hat{\beta}$ ) =\_\_\_\_\_

- A. 10.58
- B. -58.10
- C. -10.58
- D. 10.10

19. Estimate (compute) the value of slope parameter( $\hat{\beta}$ ) with zero intercept =

- A. 8.25
- B. 2.63

- C. 6.63
- D. None

## **INTERNATIONAL ECONOMICS**

### **Multiple-Choice Questions: Multiple-Choice Questions: Choose the Correct Answer**

1. An economic integration which allows reduction of tariffs, applies common external trade policies and promotes free movement of labor and capital is:
  - A. Economic union
  - B. Common market
  - C. Free trade area
  - D. Customs Union
  - E. None of the above
2. According to Monetary approach to Balance of Payments
  - A. Balance of payment disequilibrium is due to stock disequilibrium in the money market.
  - B. If currency devaluation is to improve an economy's trade balance, national output must rise relative to absorption.
  - C. The effect of a devaluation on the current balance will depend upon how it affects national income relative to how it affects domestic absorption.
  - D. Balance of payment disequilibrium can be viewed as the difference between domestic output and domestic spending
3. The WTO grew out of the following organization
  - A. The Bretton Woods system
  - B. The UN
  - C. The World Bank
  - D. The GATT
  - E. None of the above
4. Quotas are different from tariffs in that they do all of the following except.
  - A. Profits to importers, which may not contribute to net social welfare.
  - B. Rise higher degree of corruption and bribery on the part of government in case of distribution of import licenses
  - C. Correct balance of payment
  - D. Unpredictable effect of term of trade
  - E. None of the above

5. Suppose that at 2:30 Am Addis Ababa time on January 14, 2022, Ethiopian importer needed Ethiopian birr 46.789 to buy one USD. Two day a later the same importer is required Ethiopian birr 47.651 to obtain one USD. Which of the following is true about this transaction?
  - A. More dollars are obtained per Ethiopian birr on January 16, 2022.
  - B. More Birrs have to be given to obtain a dollar on January 16, 2022.
  - C. The Ethiopian birr has appreciated in value against USD.
  - D. The dollar has depreciated in values against Ethiopian birr.
6. Suppose Ethiopia is a "small" country that has no power to influence world price. Its demand and supply functions for TV sets are  $Q_d = 50 - 0.1P$  and  $Q_s = 0.1P$ , respectively. Under free trade conditions, suppose Ethiopia imports TV sets at a price of \$100 each. To protect its producers from foreign competition, suppose the Ethiopian government levies a specific tariff of \$ 100 on imported TV sets. Then, what is the reduction in Ethiopian consumer surplus due to the tariff induced in the price of TV sets?
  - A. 4500
  - B. 1000
  - C. 3500
  - D. 1500
  - E. None of the above
7. From question 5, what is the amount of deadweight welfare loss imposed on the Ethiopian economy?
  - A. 4500
  - B. 1000
  - C. 3500
  - D. 1500
  - E. None of the above
8. Heckscher–Ohlin (H-O) Theorem states that:
  - A. The immediate basis for trade derived from cost differences between nations, which were under laid by their natural and acquired advantages.
  - B. A nation will export the commodity whose production requires the intensive use of the nation's relatively abundant and cheap factor and import the commodity whose production requires the intensive use of the nation's relatively scarce and expensive factor.

- C. International trade will bring about equalization in the relative and absolute returns to homogeneous factors across nations.
  - D. A commodity is Labor intensive in the Labor abundant nation & capital intensive in the capital abundant nation.
  - E. All of the above
9. The critiques on mercantilists' view on trade, "impossible to maintain a positive trade surplus in long run", is stated by:
- A. Adam Smith
  - B. David Ricardo
  - C. David Hume
  - D. Alfred Marshall
  - E. None of the above
10. Which one of the following international trade theories advocate as commodities of equal value were exchanged, international trade considered as unproductive.
- A. The Physiocracy
  - B. The Mercantilist
  - C. The Classical
  - D. The Neoclassical
  - E. All of the above
11. Suppose that country A and country B specialize in the production of coffee and wheat which are depicted on the vertical and horizontal axes respectively, the decline cost of wheat production causes:
- A. Leftward shift in country B's offer curve
  - B. Leftward shift in country A's offer curve
  - C. Rightward shift in country B's offer curve
  - D. Rightward shift in country A's offer curve
  - E. None of the above
12. Which one of the following is not feature of the offer curve?
- A. Offer curve has positive sloped because the negative portion of the other offer curve is an irrelevant range
  - B. Offer curve has different slope
  - C. The offer curve is a composite curve of demand and supply
  - D. The offer curve is the composite curve of two outputs that a country can produce by using its resources efficiently.

E. All of the above

13. Which of the following is the most likely determinant of the terms of trade as a taxman stands ready to collect a tax based on the amounts traded between the two farmers?

A. Preferences

B. Scarcity

C. Coercion

D. Government policy

E. All of the above

The maximum amount of steel or aluminium that Canada and France can produce if they fully use all the factors of production at their disposal with the best technology available to them is shown in table 2.

Table 2: Steel and Aluminium Production

|                  | Canada | France |
|------------------|--------|--------|
| Steel (tons)     | 500    | 1200   |
| Aluminium (tons) | 1500   | 800    |

In the absence of trade, assume that Canada produces and consumes 600 tons of aluminium and 300 tons of steel and that France produces and consumes 400 tons of aluminium and 600 tons of steel.

14. If the two nations specialize based on the comparative advantage principle, by how much does specialization yield increases in output of Canada and France, respectively.

A. 600, 1000

B. 600, 200

C. 900, 1000

D. 800, 200

E. None of the above

15. Suppose Canada and France agree to a term of trade ratio 2:1 (2 ton of steel = 1 ton of aluminum). Assume that 200 tons of steel are traded for 400 tons of aluminum, by how much Canadian and French consumers better off as the result of trade, respectively?

A. 600, 1000

B. 600, 200

C. 900, 1000

D. 800, 200

E. None of the

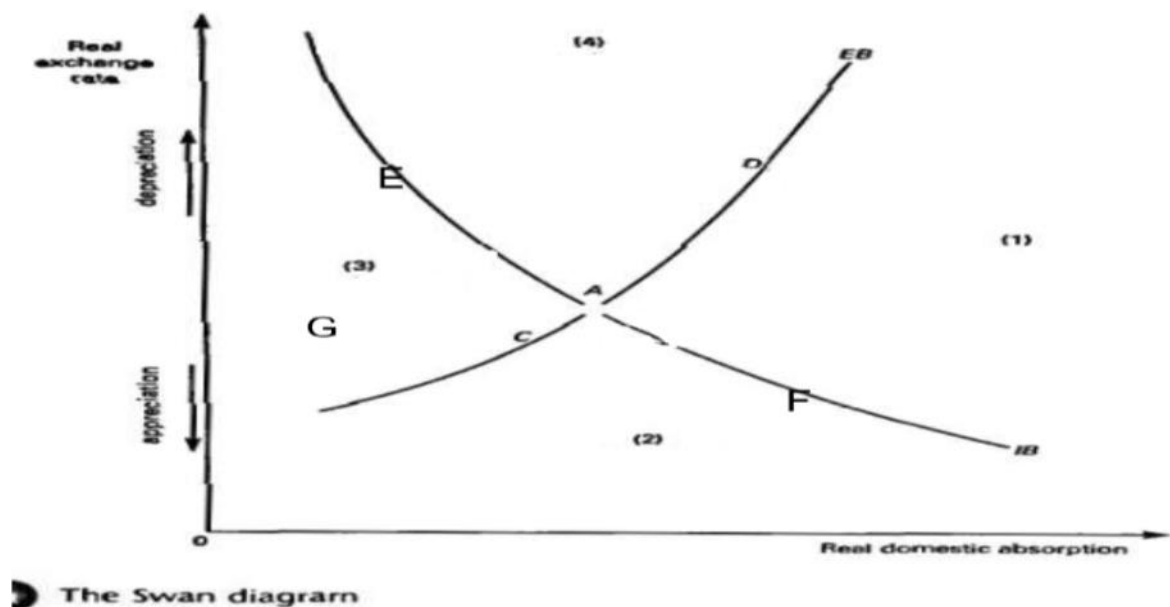
16. Import-substituting strategy of industrialization has been largely unsuccessful because of:

- A. Import substitution industries remain inefficient and costly to operate.
- B. Main beneficiaries of the import substitution process have been the foreign firms.
- C. Import substitution made possible by the heavy and often government-subsidized importation of capital goods and intermediate products.
- D. Official exchange rates have frequently been intentionally overvalued in order to boost local manufacturing by importing affordable capital and intermediate products, which in turn raises the price of exports and lowers the price of imports in terms of the local currency.
- E. All of the above

17. Accommodating items can described as

- A. A capital outflow because of higher domestic interest rate.
- B. A capital outflow to enable the financing of imports.
- C. Transactions that take place independently of the balance of payments.
- D. Increasing official reserves to finance exports.

Answer question 18 to 20 based the Swan Diagram given below



18. Point E represents one of the following possibilities;

- A. Surplus and deflation
- B. Surplus and inflation
- C. External balance

- D. Internal balance
19. Suppose, the economy is at point G, what is (are) the appropriate policy measure(s) the authorities should take to tackle the economic problems the economy experiences at point G?
- Increasing real domestic expenditure
  - Revaluating the exchange rate
  - Increasing real domestic expenditure and revaluating the exchange rate
  - Cutting real domestic expenditure and devaluing the exchange rate
20. Zone 1 depicts:
- A deficit and unemployment pressures
  - A deficit and inflationary pressures
  - A surplus and unemployment pressures
  - A surplus and inflationary pressures

## MATHEMATICAL ECONOMICS

### Choose the best answer for the following multiple questions

- Which of following functions are strictly increasing at  $x = 4$ .
  - $y = 3x^2 - 14x + 5$
  - $y = x^3 - 7x^2 + 6x - 2$
  - $y = x^4 - 6x^3 + 4x^2 - 13$
  - none
- If the demand function is given as  $q = \sqrt{10 - p^2}$ , then the price elasticity of demand at  $p = 2$  is;
  - $-2/3$
  - $-4/6$
  - $-2/7$
  - a and b
- A firm knows that the demand function for its output is  $P = 400 - 0.5q$ . What price should it charge to maximize sales revenue?
  - 200
  - 400
  - 250
  - None
- When will average variable cost be at its minimum value for the TC function,  $TC = 40 + 82q - 6q^2 + 0.2q^3$ ?
  - 233
  - 555
  - 217
  - 15

5. A firm faces the production function  $Q = 120L + 200K - L^2 - 2K^2$  for positive values of  $Q$ . It can buy  $L$  at Birr5 a unit and  $K$  at Birr8 a unit and has a budget of Birr70. What is the maximum output it can produce?
- $L = 38.57$ ;  $K = 5.79$
  - $L = 7.74$ ;  $K = 5.79$
  - $L = 38.57$ ;  $K = 32.86$
  - $L = 7.74$ ;  $K = 32.26$

*Solve the following three sector comparative statistic and answer question number 6-8*

$$\begin{array}{lllll} Y = C + I_0 + G_0 & Yd = Y - T & C_0 = 100 & I_0 = 90 & b = 0.75 \\ C = C_0 + bYd & T = T_0 + ty & G_0 = 330 & T_0 = 240 & t = 0.20 \end{array}$$

6. The equilibrium level of income is 850
- True
  - False
7. When government spending increased by Birr 50,
- Equilibrium income increased by 500
  - Equilibrium income decreased by 250
  - Equilibrium income remain the same
  - none
8. As autonomous taxation ( $T_0$ ) increased by 50, then equilibrium income increased by 93.75
- True
  - False
9. for a function  $y = -2x^3 + 4x^2 + 9x - 15$  the rate of change at  $x=3$ , will be
- Increasing at decreasing rate and concave
  - Decreasing at decreasing rate and concave
  - Increasing at increasing rate and convex
  - Decreasing at increasing rate and convex
10. Given the demand for Teff;  $Q_T = 4850 - 5P_T + 1.5P_W + 0.1Y$ , with  $Y = 10000$ ,  $P_T = 200$ , and the price for Wheat  $P_W = 100$ . The find income elasticity and cross price elasticity of Teff.
- The income elasticity is 0.2 and cross elasticity is 0.03
  - The income elasticity is 0.03 and cross elasticity is 0.02
  - The income elasticity is 0.2 and cross elasticity is 0.3
  - The income elasticity is 0.3 and cross elasticity is 0.2
11. A function  $f(x)$  is concave at  $x = a$ , if
- $f''(a) > 0$
  - $f''(a) < 0$
  - $f''(a) = 0$
  - $f'(a) < 0$
12. Which one of the following techniques of estimation predicts more accurate than others.
- Linear approximation
  - Quadratic approximation

- C. Polynomial approximation  
D. None
13. For a production function  $Q(x)$ , the maximum point at  $x = a$  is characterized as;
- A.  $Q'(a) = 0$  and  $Q''(a) < 0$   
B.  $Q'(a) = 0$  and  $Q''(a) > 0$   
C.  $Q'(a) < 0$  and  $Q''(a) < 0$   
D.  $Q'(a) > 0$  and  $Q''(a) = 0$
14. Suppose  $f(x)$  is continuous for all values of  $x$  in the closed interval  $[a, b]$ , then IVT theorem states there is
- A. Exactly one solution  
B. Many positive solution  
C. Only Two solution  
D. None
15. Given a Cobb- Douglas production function  $Q(L, K) = 60L^{3/4}K^{1/4}$ , identify the odd
- A. The function is homogeneous of degree zero  
B. The first order partial derivative with respect to each input variables are homogenous of degree zero  
C. The firm is operating at decreasing returns to scale  
D. All
16. Which one of the following represents third- order and fifth degree differential equations
- A.  $\left(\frac{dy}{dt}\right)^3 + 5t = 0$   
B.  $\left(\frac{dy}{dt}\right)^3 + 2\left(\frac{dy}{dt}\right)^5 + y = 2$   
C.  $\left(\frac{d^5y}{dt^5}\right)^3 = 4y$   
D.  $\left(\frac{d^3y}{dt^3}\right)^5 - 3y = 40$
17. The particular integral equation for  $y''(t) + y'(t) = -10$  will be
- A.  $Y_p = -10$   
B.  $Y_p = -10t$   
C.  $Y_p = -10t^2$   
D. None
18. The complementary solution for Q#17 is expressed as;
- A.  $Y_c = A_1 e^{-t} + A_2 e^{-t}$   
B.  $Y_c = A_1 + A_2 e^{-t}$   
C.  $Y_c = A_1 e^{-0.5t} + A_2 e^{-0.75t}$   
D. None
19. One of the following differential equation is not exact
- A.  $(4y + 8t^2)dy + (16yt - 3)dt = 0$   
B.  $(2yt^3)dy + (3y^2t^2)dt = 0$   
C.  $8tyy' = -(3t^2 + 4y^2)$

D.  $\frac{dy}{dt} + \frac{2y^4t+3t^3}{4y^3t^2} = 0$

20. The definite solution for the first order difference equation given by  $y_{t+1} - 5y_t = 1$  at  $y_0 = 7/4$  is
- A.  $Y_t = 2(5)^t + 1/4$
  - B.  $Y_t = A(5)^t - 1/4$
  - C.  $Y_t = 2(5)^t - 1/4$
  - D.  $Y_t = A(5)^t + 1/4$

## DEVELOPMENT ECONOMICS

Part I: Choose the best answer and round it (1 pts each)

1. Economic development refers to \_\_\_\_\_.
  - A. Economic growth.
  - B. Economic growth plus changes in output distribution and economic structure.
  - C. Improvement in the well-being of the urban population.
  - D. Sustainable increases in Gross National Product
  - E. All of the above are correct
2. Among the following determinants of growth, which is a non-economic factor?
  - A. Natural resources
  - B. Population growth
  - C. Favourable legislation
  - D. Capital accumulation
  - E. All of the above
3. One of the following is an economic indicator of development
  - A. Low rate of population growth
  - B. Increased per capita income
  - C. Democratic administration
  - D. Well provision of education service
  - E. All
4. A period of rapid population growth between a preindustrial, stable population characterized by high birth and death rates and a later, modern, stable population marked by low fertility and mortality is known as
  - A. demographic transition.
  - B. population maturity.
  - C. demobilizing population.
  - D. birth-death transformation.

Imagine that a country has a national savings rate of 20% and that the capital-output ratio is 4. Finally, suppose that capital lives forever. (Assuming the Harrod-Domar model)

5. What the capital-output ratio should be to get a growth rate of 10% per year.
- A. 2
  - B. 0.5
  - C. 1
  - D. None
6. If the distribution of income for country X with a total population of six individuals is given as (1, 1.5, 2, 3, 5, 5) and the poverty line is \$1.9, the headcount measure tells us what percentage of the population is in an absolute poverty situation

10%

- A. 1%
- B. 26.67%
- C. 33.33%
- D. None

Assume that the current PCI for developed ( $YD_t$ ) = 20,000; for developing ( $YDC_t$ ) = 650; per capita income growth for developed ( $rD$ ) for a period (2000-2020) = 2 %; current per capita income growth for a developing country ( $rDC$ ) = 7.5%.

7. Given current PCI growth how long would it take LDCs to reach the current average PCI of the DCs? And interpret the result.
- A. 20 years
  - B. 23 years
  - C. 41 years
  - D. None
8. Which stage of growth of Rostow is called a brake through
- A. The traditional society
  - B. Pre-condition for the take-off into self-sustaining growth
  - C. Take-off stage
  - D. Drive to maturity
  - E. The age of high consumption-stages of self-sustained growth
9. One of the following is odd
- A. Lewis development theory
  - B. The Harrod-Domar growth model
  - C. The Solow growth model

- D. Rostow's stages of growth model
10. Disguised unemployment is a situation where:
- A. people who are willing to work are unable to find jobs
  - B. people in the labor force hold two or more jobs
  - C. people are working full-time and hard, yet have a low marginal product
  - D. people work in a free labor market with a very low wage
  - E. none of the above
11. According to Simon Kuznet, the relationship between GNP per capita and inequality in the distribution of income can be expressed as:
- A. a strictly decreasing relationship
  - B. a strictly increasing relationship
  - C. no relationship
  - D. first decreasing and then increasing
  - E. none of the above
12. The Gini coefficient provides a measure of:
- A. the level of poverty
  - B. the level of relative inequality
  - C. disguised unemployment
  - D. the rate of growth
  - E. none of the above
13. Balanced growth theories of Ragnar Nurkse advocate
- A. a steady rate of growth of GNP
  - B. parity between capital stock and labor
  - C. simultaneous development in all sectors of the economy
  - D. mechanization of agriculture
  - E. All of the above.
14. Economic growth in one area adversely affects the prosperity of another refers,
- A. Spread effect
  - B. Backwash effect
  - C. trickle-down economics effect
  - D. pro-growth effects
  - E. none of the above
15. Which of the following does not constitute economic development in 'the new view'?
- A. Improvement in material welfare

- B. Mere increase in the real GDP
  - C. Quantitative and qualitative changes in life
  - D. Production and employment decisions open to all citizens
  - E. none of the above
16. What are the causes of lower agricultural performance of sub-Saharan African relative to other LDCs?
- A. Infeasibility of traditional agricultural practices.
  - B. Deterioration of soil nutrients.
  - C. Farmers' inability to use modern agricultural inputs.
  - D. All of the above.
17. Which of the following is a not form of urban bias?
- A. Tax incentives and subsidies to the infant-industry.
  - B. Tariff and quota protection for industry.
  - C. Spending more for education, training, housing, health and transport in urban areas than in rural areas.
  - D. Setting price floors on food and price ceilings for industrial goods.
  - E. None of the above.
18. Which of the following statement incorrectly explains state of LDCs international trade?
- A. Many developing countries rely heavily on exports of primary products.
  - B. Many developing countries have chronic deficits on balance of payment.
  - C. Many developing countries rely heavily on export of capital goods.
  - D. None of the above.
19. Prebisch-Singer hypothesis states that;
- A. Primary commodity export prices rise and secondary commodity export price falls over time.
  - B. Developing countries need to diversify their export.
  - C. Developing countries need to depend on primary exports.
  - D. All of the above.
20. What is/are the motivation/s of the foreign aid?
- A. To control strategic locations for military and economic purpose.
  - B. To solve foreign exchange constraint.
  - C. To fill saving gap.
  - D. All of the above

